

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	18 June 2026
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Project Name	ShopEZ – Full Stack E-Commerce Platform Using MERN Stack
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through form using name, email, and password
FR-2	User Login	Login using registered email and password
FR-3	User Authentication	Secure access using authentication/token system
FR-4	User Profile	View and update user profile details
FR-5	Product Listing	Display all available products to users
FR-6	Product Details	View product name, price, image, description, and category
FR-7	Cart Management	Add product to cart, update quantity, remove product, and view total amount
FR-8	Order Management	Place order, view order history, and check order status
FR-9	Admin Login	Admin can log in securely to access dashboard
FR-10	Admin Product Management	Admin can add, update, and delete products
FR-11	Admin Order Management	Admin can view and manage customer orders
FR-12	Database Management	Store users, products, cart, and orders in database collections

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application should be simple, clear, and easy to use for both users and admin.
NFR-2	Security	User login details and data should be protected using secure authentication.
NFR-3	Reliability	The application should work correctly without errors during product browsing, cart usage, and order placement.

NFR-4	Performance	Pages should load quickly and respond fast to user actions.
NFR-5	Availability	The application should be available whenever the user or admin needs to access it.
NFR-6	Scalability	The system should support future features like payment gateway, tracking, and mobile app support.
NFR-7	Maintainability	The frontend, backend, and database code should be organized so that future changes can be made easily.
NFR-8	Compatibility	The application should work properly on common browsers and different screen sizes.